Prem Vyas

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EDUCATION

University of Alberta **Sep 2025 - April 2030**
BSc Mechatronics & Robotics Engineering Co-op Edmonton, AB

EXPERIENCE

Undergraduate Research Assistant **July 2026 – Present**
Neuromuscular Control & Biomechanics Laboratory Edmonton, AB

- Developing a 5-DOF cable-driven hand exoskeleton for children with cerebral palsy, actuated from a backpack unit to keep hand-mounted mass under 200 g.
- Derived a method generating 25 CAD dimensions from a single hand-length measurement, validated against published pediatric data at 6% mean absolute error
- Defined three device sizes covering ages 6–18 and selected the finger drive mechanism from four candidate designs.

Mechanical Team Member **May 2026 – Present**
University of Alberta Robotics and Aerial Drones (UA-RAD) Edmonton, AB

- Building a payload-dropper mechanism in SolidWorks for a hexacopter to hold and release a 10–12 kg payload, using a self-locking ratchet and pawl design
- Designing motor mounts sized to react up to 8.4 kg of thrust, clamping onto a carbon-fiber arm without crushing the tube.
- Ran physical load tests on carbon-fiber arms, sizing the test load from motor thrust data and a safety factor to verify the arms survive flight loads without failing.

Aerodynamics Team Member **Feb 2026 – Present**
University of Alberta Formula Racing Edmonton, AB

- Designed a low-drag single-surface radiator cooling scoop that delivers airflow to the radiator face with near-zero pressure loss and near-uniform distribution at 15 m/s.
- Running coastdown simulations in RaceStudio 3 to evaluate aerodynamic drag data and improve the aerodynamic performance of the Formula Racing car.
- Performing Computational Fluid Dynamics (CFD) simulations using Ansys to analyze airflow patterns, drag coefficients, and pressure distribution.

PROJECTS

STM32 Flight Controller | *ArduPilot, KiCad, Python* **March 2026 – June 2026**

- Designed a custom flight controller PCB featuring an STM32 microcontroller, a BMP580 barometer for altitude estimation, and microSD data logging.
- Built a dual-input power management system using buck and buck-boost regulators to ensure stable 3.3V and 5V power delivery.

Robotic Rock-Paper-Scissors Hand | *SolidWorks, C++ , Python, OpenCV, MediaPipe* **June 2026 – Present**

- Building a 3D-printed tendon-driven robotic hand actuated by servos, controlled by a Ardiuno, to play rock paper scissors against a person.
- Developing computer-vision gesture detection to read the player's hand in real time, paired with a move prediction strategy to choose the counter-throw.

TECHNICAL SKILLS

CAD & Simulation | SolidWorks, Onshape, Ansys (CFD), RaceStudio 3, LoggerPro
Electrical & Embedded | *KiCad*, ArduPilot, STM32, Arduino, Power Electronics
Programming | Python, Visual Basic, C/C++, Linux/CLI, OpenCV, MediaPipe, Git
Manufacturing| 3D Printing (FDM), CO_2 Laser Cutter, Fibre Cutter, Waterjet Cutting, Hand and Power Tools